Customer Churn Prediction Using Supervised Learning Models

Nidhin G · Reg. No. 2117240030094 · Class: AIML-B

Department of Artificial Intelligence & Machine Learning

Rajalakshmi Institute of Technology

INTRODUCTION

Understanding Customer Churn

What Is Churn?

Customer churn occurs when a subscriber discontinues a service — a critical metric for telecom, SaaS, and subscription businesses.

Why Predict It?

Early identification of at-risk customers enables proactive retention strategies before they leave.

Business Impact

Retaining an existing customer costs up to **5× less** than acquiring a new one. Churn reduction directly improves revenue.

PROBLEM STATEMENT

Problem Statement & Project Goals

The Core Challenge

Predict whether a customer will churn (Yes/No) using supervised learning — while addressing class imbalance, selecting the right decision threshold, and maximizing actionable accuracy.

Class imbalance skews model learning toward the majority class

Standard **accuracy** is misleading with imbalanced data

Optimal **decision threshold** must be tuned per model

Project Goals

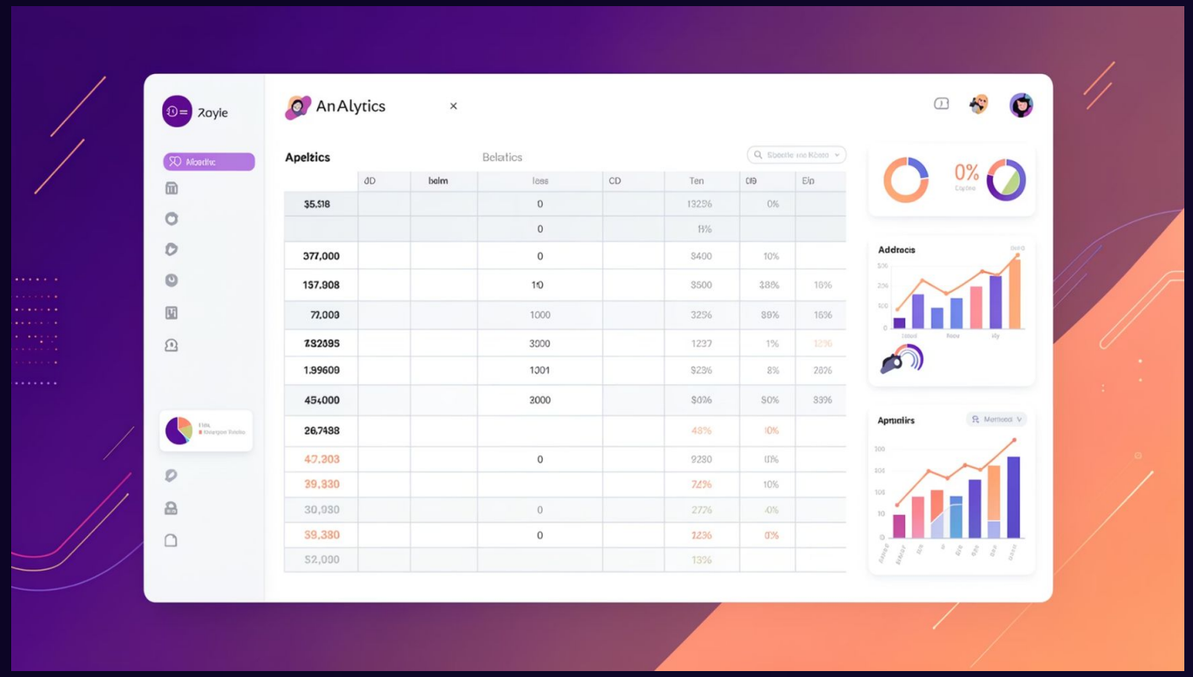
- Build accurate churn prediction models

Handle class imbalance using **SMOTE**

- Optimize classification threshold per model
- Deliver actionable business risk insights

DATASET

Telco Customer Churn Dataset



The **Telco Customer Churn** dataset from Kaggle is a widely-used benchmark for binary classification in churn prediction research.

~7,043 Customers

Real-world telecom subscriber records

21 Features

Tenure, charges, contract type, payment method, demographics

Binary Target

Churn: **Yes** or **No** (26% positive class)

METHODOLOGY

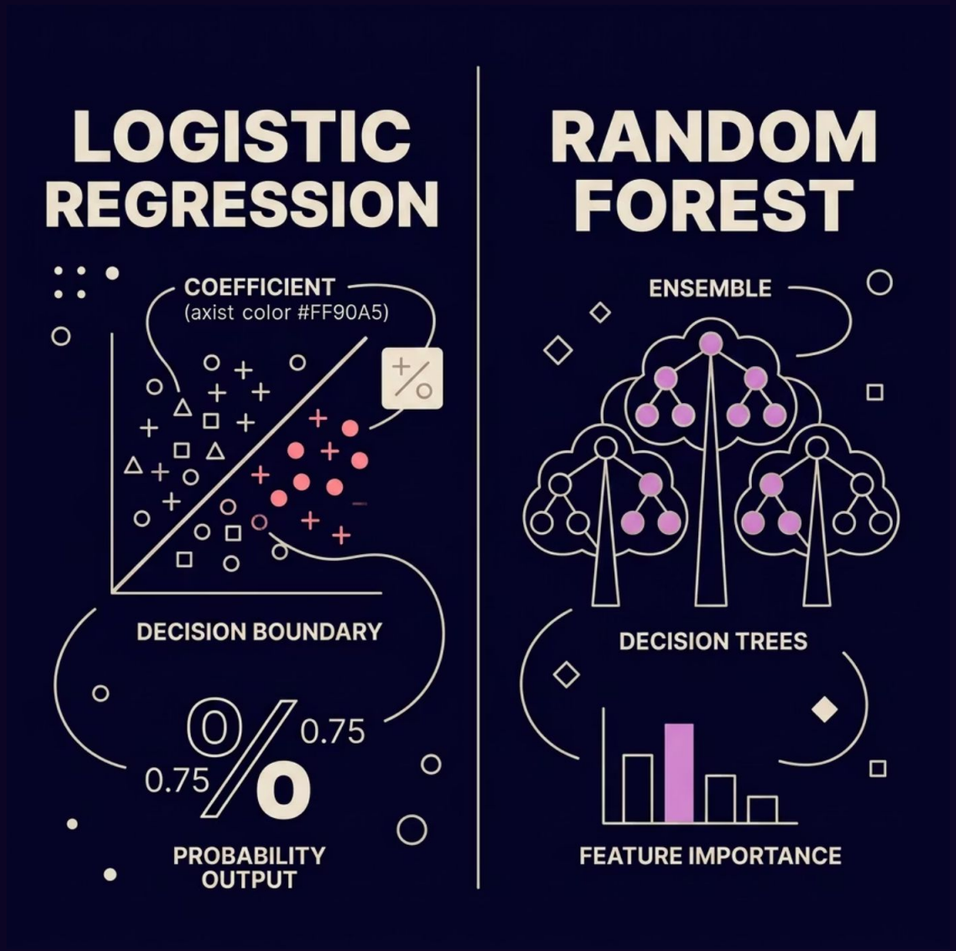
End-to-End Project Workflow

Preprocess → **Handle Imbalance** → **Train Models** → **Tune Threshold** → **Evaluate Insights**

Each stage is designed to maximize predictive performance while ensuring the model generalizes well to unseen customer data.

ALGORITHMS

Models Under the Hood



Logistic Regression

A linear classifier that models the probability of churn. Highly interpretable — coefficients reveal feature impact directly.

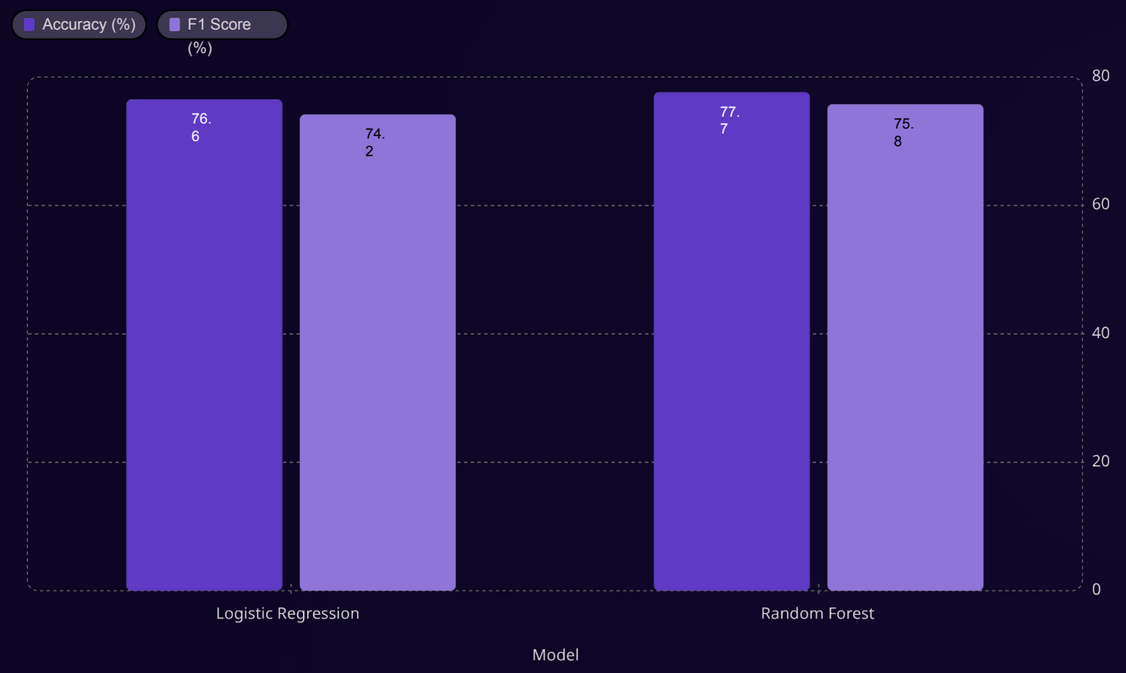
Random Forest

An ensemble of decision trees that reduces overfitting through bagging. Captures complex, non-linear patterns in customer behavior for stronger generalization.

> Both models output probability scores, enabling custom threshold tuning for recall optimization.

RESULTS

Model Performance Comparison



Key Findings

Random Forest Wins

77.68% accuracy — a **1.13 pp improvement** over Logistic Regression

F1 Score Matters

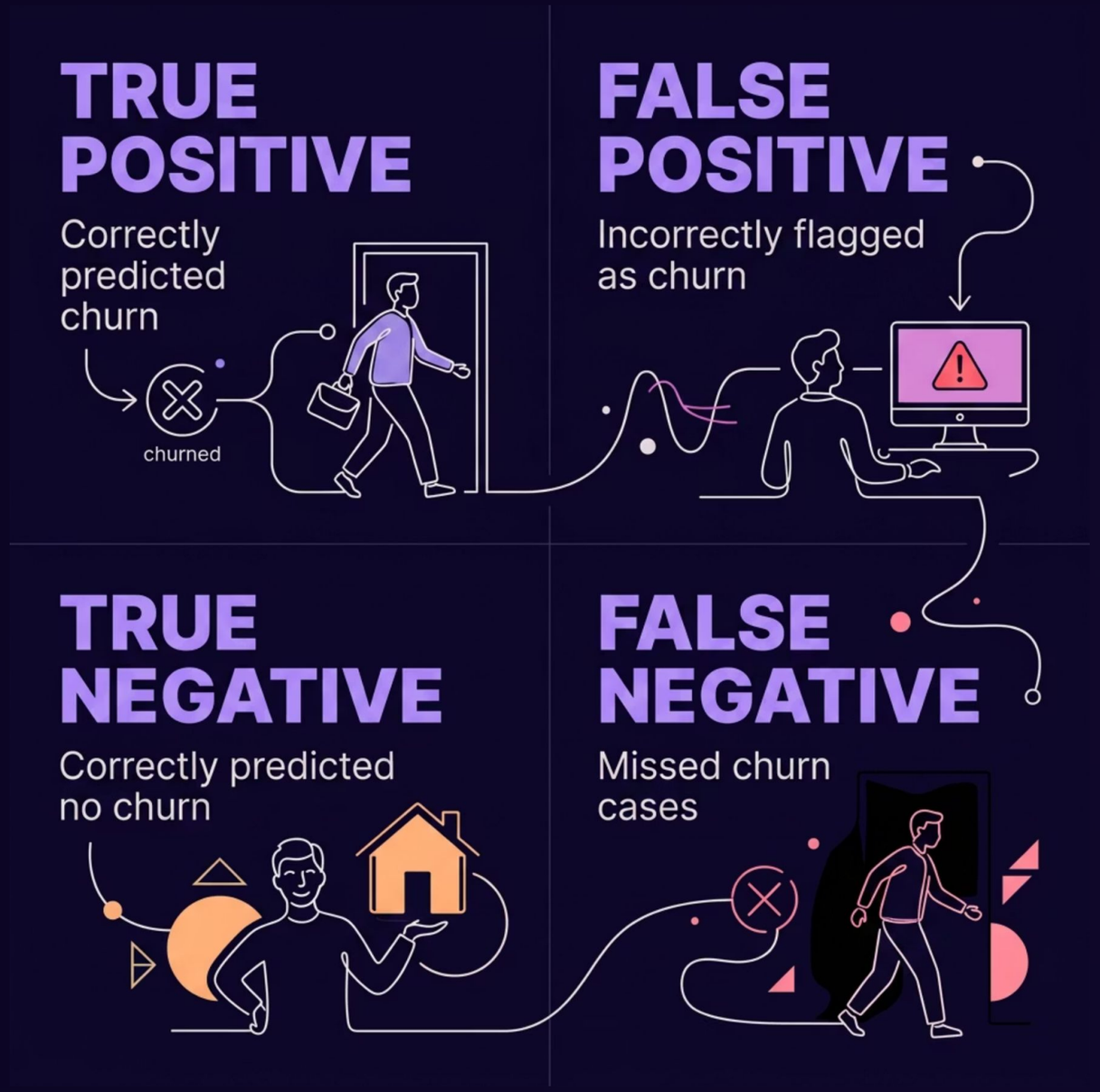
Balances precision and recall — critical when churn is the minority class

SMOTE Impact

Synthetic oversampling improved minority class representation and recall

EVALUATION

Confusion Matrix & Threshold Tuning



Why Threshold Tuning?

The default 0.5 threshold may miss churners. Lowering it increases recall — catching more at-risk customers.

⚠ **FN (missed churners)** is the costliest error — threshold tuning directly targets this.

LR Threshold: 0.49

RF Threshold: 0.39

BUSINESS INSIGHTS

Risk Segmentation for Action

🟢 Low Risk

~9% Churn Rate Stable customers — maintain engagement with standard service

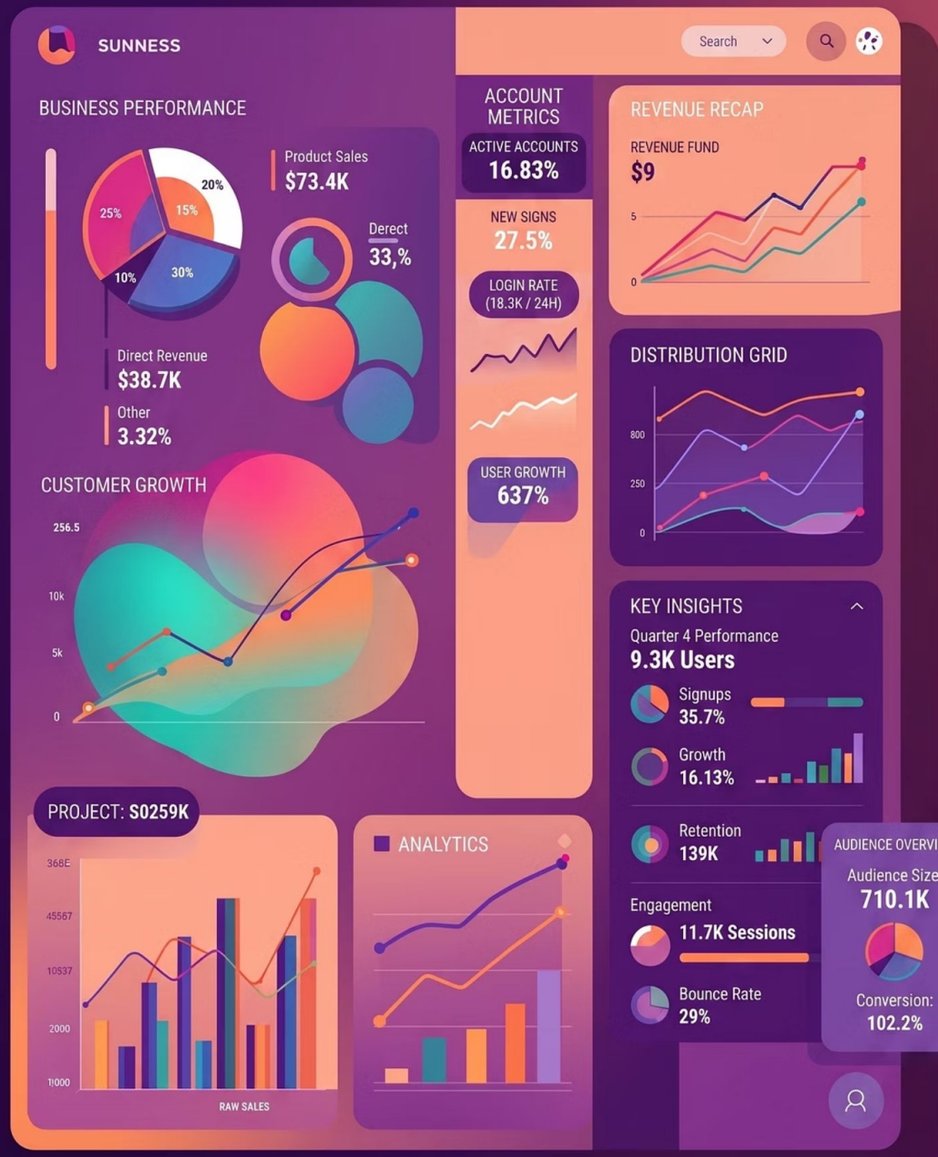
🟡 Medium Risk

~33% Churn Rate Proactive outreach recommended — offers, loyalty programs

🔴 High Risk

~60% Churn Rate Immediate intervention required — dedicated retention team

Key Insight: High-risk customers represent the highest ROI for retention campaigns. Prioritize them first.



CONCLUSION

Conclusion & Future Directions

What We Achieved

- **Random Forest** outperformed Logistic Regression (77.68% vs 76.55% accuracy)
- **SMOTE** effectively addressed class imbalance, improving minority recall
- **Threshold tuning** optimized recall — reducing costly missed churners

Future Work

XGBoost / Gradient Boosting for stronger ensemble performance

Web Application deployment for real-time churn scoring

Real-time prediction pipeline integrated with CRM systems

Explainable AI (SHAP) for transparent business decisions